LAB-ASSIGNMENT-3

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ENROLLMENT:-S25MCAG0010

1). **Display the details of students sorted by age in ascending order.**

```
mysql> SELECT * FROM Students
    -> ORDER BY Age ASC;
+-----------------+----------+----------+-------------+------+---------------+
| Registration_id | S_name   | Semester | Course_code | Age  | Instructor_id |
+-----------------+----------+----------+-------------+------+---------------+
| SCS090          | Shiva    |        3 | CS303       |   16 |           104 |
| SCS012          | Arjun    |        3 | CS301       |   17 |           101 |
| SCS015          | Abhishek |        3 | CS303       |   17 |           104 |
| SCS013          | Anamika  |        3 | CS301       |   19 |           101 |
| SCS014          | Diya     |        1 | CS101       |   19 |           103 |
| SCS016          | Divya    |        3 | CS302       |   20 |           102 |
| SCS022          | Alam     |        5 | MBA501      |   21 |           108 |
| SCS071          | Rahul    |        5 | CS505       |   21 |           105 |
| SCS043          | Rupam    |        3 | EC303       |   22 |           107 |
| SCS055          | Ankit    |        5 | MCA         |   22 |           103 |
| SCS017          | Hina     |        3 | EC303       |   23 |           107 |
| SCS056          | Satya    |        7 | CS701       |   23 |           106 |
+-----------------+----------+----------+-------------+------+---------------+
12 rows in set (0.00 sec)
```

2). **Display the number of students in each semester.**

```
mysql> SELECT Semester, COUNT(*)
    -> FROM Students
    -> GROUP BY Semester;
+----------+----------+
| Semester | COUNT(*) |
+----------+----------+
|        3 |        7 |
|        1 |        1 |
|        5 |        3 |
|        7 |        1 |
+----------+----------+
4 rows in set (0.01 sec)
```

3). Display the courses which are having more than 2 credits sorted in descending order.

```
mysql> SELECT * FROM Courses
    -> WHERE Credits > 2
    -> ORDER BY Credits DESC;
+-------------+-----------------+---------+----------+---------------+
| Course_code | Course_title    | Credits | Semester | Instructor_id |
+-------------+-----------------+---------+----------+---------------+
| CS301       | DBMS            |       4 |        3 |           101 |
| CS505       | Discrete Math   |       4 |        5 |           105 |
| CS302       | OS              |       3 |        3 |           102 |
| CS101       | Physics         |       3 |        1 |           103 |
| CS303       | Java            |       3 |        3 |           104 |
| CS701       | Cloud Computing |       3 |        7 |           106 |
| MBA501      | Economics       |       3 |        5 |           108 |
+-------------+-----------------+---------+----------+---------------+
7 rows in set (0.03 sec)
```

4). Display courses offered in semesters 3 and 5 ordered by course code.

```
mysql> SELECT * FROM Courses
    -> WHERE Semester IN (3,5)
    -> ORDER BY Course_code;
+-------------+---------------+---------+----------+---------------+
| Course_code | Course_title  | Credits | Semester | Instructor_id |
+-------------+---------------+---------+----------+---------------+
| CS301       | DBMS          |       4 |        3 |           101 |
| CS302       | OS            |       3 |        3 |           102 |
| CS303       | Java          |       3 |        3 |           104 |
| CS505       | Discrete Math |       4 |        5 |           105 |
| EC303       | DLD           |       2 |        5 |           107 |
| MBA501      | Economics     |       3 |        5 |           108 |
+-------------+---------------+---------+----------+---------------+
6 rows in set (0.01 sec)
```

5). List courses by descending order of credits and then ascending order of semesters.

```
mysql> SELECT * FROM Courses
    -> ORDER BY Credits DESC, Semester ASC;
+-------------+-----------------+---------+----------+---------------+
| Course_code | Course_title    | Credits | Semester | Instructor_id |
+-------------+-----------------+---------+----------+---------------+
| CS301       | DBMS            |       4 |        3 |           101 |
| CS505       | Discrete Math   |       4 |        5 |           105 |
| CS101       | Physics         |       3 |        1 |           103 |
| CS302       | OS              |       3 |        3 |           102 |
| CS303       | Java            |       3 |        3 |           104 |
| MBA501      | Economics       |       3 |        5 |           108 |
| CS701       | Cloud Computing |       3 |        7 |           106 |
| EC303       | DLD             |       2 |        5 |           107 |
+-------------+-----------------+---------+----------+---------------+
8 rows in set (0.01 sec)
```

6). Find the maximum credits among courses for each instructor.

```
mysql> SELECT Instructor_id, MAX(Credits)
    -> FROM Courses
    -> GROUP BY Instructor_id;
+---------------+--------------+
| Instructor_id | MAX(Credits) |
+---------------+--------------+
|           101 |            4 |
|           102 |            3 |
|           103 |            3 |
|           104 |            3 |
|           105 |            4 |
|           106 |            3 |
|           107 |            2 |
|           108 |            3 |
+---------------+--------------+
8 rows in set (0.01 sec)
```

7). Display the number of courses offered by each instructor in semester 3.

```
mysql> SELECT Instructor_id, COUNT(*)
    -> FROM Courses
    -> WHERE Semester = 3
    -> GROUP BY Instructor_id;
+---------------+----------+
| Instructor_id | COUNT(*) |
+---------------+----------+
|           101 |        1 |
|           102 |        1 |
|           104 |        1 |
+---------------+----------+
3 rows in set (0.00 sec)
```

8). Create a view that lists instructors and the number of courses they teach.

```
mysql> CREATE VIEW Instructor_Course_Count AS
    -> SELECT Instructor_id, COUNT(*) AS Total_Courses
    -> FROM Courses
    -> GROUP BY Instructor_id;
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> SELECT * FROM Instructor_Course_Count;
+---------------+---------------+
| Instructor_id | Total_Courses |
+---------------+---------------+
|           101 |             1 |
|           102 |             1 |
|           103 |             1 |
|           104 |             1 |
|           105 |             1 |
|           106 |             1 |
|           107 |             1 |
|           108 |             1 |
+---------------+---------------+
8 rows in set (0.01 sec)
```

9). Create a view that lists students and the number of courses they are registered for.

```
mysql> CREATE VIEW Student_Course_Count AS
    -> SELECT Registration_id, COUNT(Course_code) AS Total_Courses
    -> FROM Students
    -> GROUP BY Registration_id;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> SELECT * FROM Student_Course_Count;
+-----------------+---------------+
| Registration_id | Total_Courses |
+-----------------+---------------+
| SCS012          |             1 |
| SCS013          |             1 |
| SCS014          |             1 |
| SCS015          |             1 |
| SCS016          |             1 |
| SCS017          |             1 |
| SCS022          |             1 |
| SCS043          |             1 |
| SCS055          |             1 |
| SCS056          |             1 |
| SCS071          |             1 |
| SCS090          |             1 |
+-----------------+---------------+
12 rows in set (0.02 sec)
```

10). Create a view that shows students registered in semester 3, ordered by age.

```
mysql> CREATE VIEW Semester3_Students AS
    -> SELECT * FROM Students
    -> WHERE Semester = 3
    -> ORDER BY Age;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SELECT * FROM Semester3_Students;
+-----------------+----------+----------+-------------+------+---------------+
| Registration_id | S_name   | Semester | Course_code | Age  | Instructor_id |
+-----------------+----------+----------+-------------+------+---------------+
| SCS090          | Shiva    |        3 | CS303       |   16 |           104 |
| SCS012          | Arjun    |        3 | CS301       |   17 |           101 |
| SCS015          | Abhishek |        3 | CS303       |   17 |           104 |
| SCS013          | Anamika  |        3 | CS301       |   19 |           101 |
| SCS016          | Divya    |        3 | CS302       |   20 |           102 |
| SCS043          | Rupam    |        3 | EC303       |   22 |           107 |
| SCS017          | Hina     |        3 | EC303       |   23 |           107 |
+-----------------+----------+----------+-------------+------+---------------+
7 rows in set (0.01 sec)
```

11). Create a view that shows the count of instructors in each department, ordered by department name.

```
mysql> CREATE VIEW Department_Instructor_Count AS
    -> SELECT Department, COUNT(*) AS Total_Instructors
    -> FROM Instructors
    -> GROUP BY Department
    -> ORDER BY Department;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> SELECT * FROM Department_Instructor_Count;
+------------+-------------------+
| Department | Total_Instructors |
+------------+-------------------+
| CSE        |                 5 |
| EC         |                 1 |
| MBA        |                 1 |
| ME         |                 1 |
+------------+-------------------+
4 rows in set (0.01 sec)
```